

Macroeconomic Theory and Policy: Lecture W3 (Ch.9)

What are You Going to Learn (Ch. 9)

- How do we measure the economy's performance in the short run?
 - ▶ Using the gap between actual GDP and potential GDP
- How costly fluctuations in economic activity can be
- The rate of inflation tends to decline when the economy is in a recession
- A simple version of the short-run model that will help us understand these patterns

The Long Run, the Short Run, and Shocks

- The long-run model describe how the economy behaves on average
 - ▶ Solow model describes the steady state
 - ▶ Romer model describes the long run growth rate
 - ▶ The quantity theory of money sets long-run inflation
- Potential output: amount of production if all inputs were utilized at their long-run sustainable levels
- At any given time, the economy is unlikely to exactly equal the long-run average
 - ▶ The short run can be thought of as the length of time over which these deviations occur
 - ▶ Deviations typically last for two years or so

The Long Run, the Short Run, and Shocks

In the short-run model,

- the current level of output and inflation are endogenous
- assume that the long run variables are given exogenous
 - ▶ Potential output and the long-run rate of inflation are determined by the long-run model
 - ▶ Since these variables are outside the short-run model, they are exogenous
- current output can be deviated from potential output because of exogenous economic shocks

Trends and Fluctuations

- Actual output is equal to the long-run trend plus short-run fluctuations

Actual Output = Long-run trend + short-run fluctuations

$$Y_t = \bar{Y}_t + \bar{Y}_t \times \tilde{Y}_t$$

where

$\bar{Y}_t$ is the long-run trend (potential output) in level

$\tilde{Y}_t$ is the *percentage* deviations from $\bar{Y}$ representing the short-run fluctuations

Short-Run Fluctuation

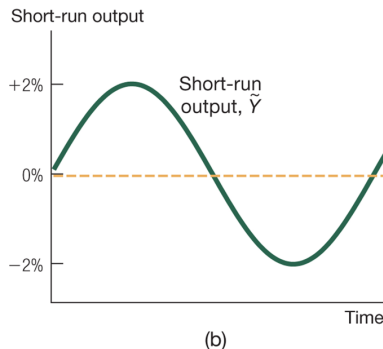
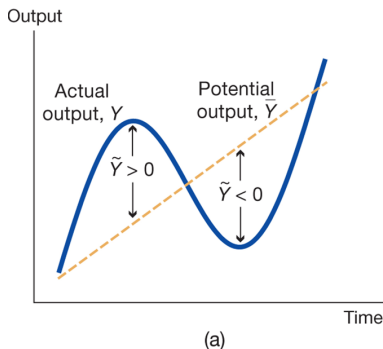
- “Detrended output” or short-run output is the difference in actual and potential output, expressed as a percentage of potential output:

$$\tilde{Y}_t = \frac{Y_t - \bar{Y}_t}{\bar{Y}_t}$$

- Why do we express it in percentage?
 - ▶ Suppose actual output falls short of potential output by \$100 million in a given year
 - ▶ It depends on the overall size of the economy. In the US, \$100 million would be a much smaller fraction of GDP than is in Canada
 - ▶ It is more informative to say actual output is, for example, 1 percent below potential
 - ▶ This way, we can compare short-run measure across time and location

Economic Fluctuations and Short-Run Output

- Panel (a) shows actual and potential output
- Panel (b) shows the implied short-run fluctuations
- When we remove the long-run trend associated with economic growth (potential output), what remains are the ups and downs of economic fluctuations
- When there is an expansion (recession), and the economy is booming (contracting), actual output is above (below) potential, and $\tilde{Y}$ is positive (negative)

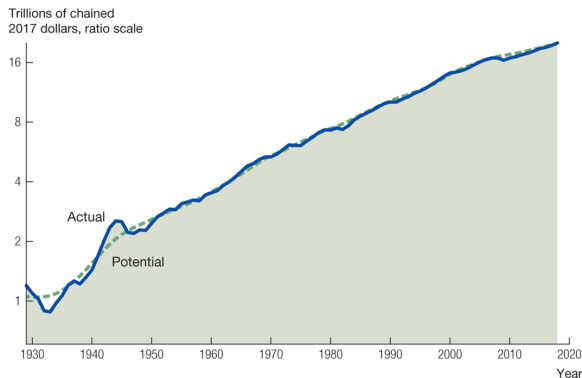


Short-Run Output in the United States

- Fluctuations in U.S. GDP:
 - ▶ Difficult to see graphically over a long period of time if we plot it at level
 - ▶ Have mostly been between + or -4% since 1950
- The Great Depression:
 - ▶ Negative gap during the 1930s
 - ▶ Actual output was well below potential output
 - ▶ Take a long time to recover

U.S. Real GDP, Actual and Potential

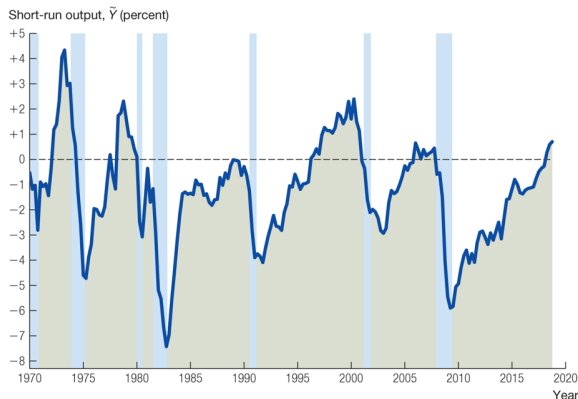
- Because of the magnitude of economic growth, actual and potential output look quite similar on this time scale, even in ratio scale
- If we compute the difference between actual and potential output, the economic fluctuations become visible



Source: U.S. Department of Commerce. Potential output is constructed by the Congressional Budget Office for years after 1949.
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U.S. Economic Fluctuations

- A recession begins when actual output falls below potential and short-run output becomes negative
- A recession usually ends when short-run output starts to rise and become less negative before output returns to potential



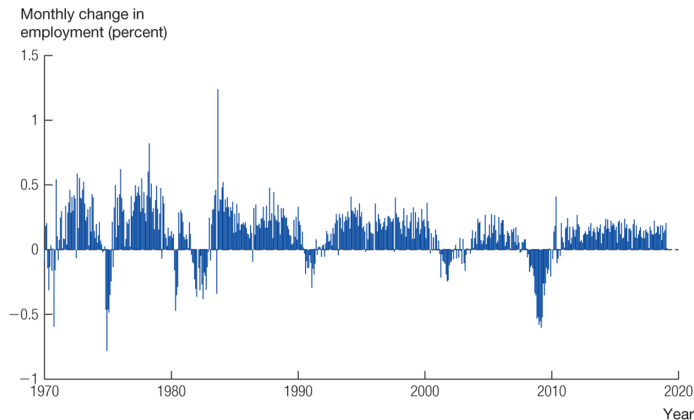
Source: Federal Reserve Economic Data (FRED), courtesy of the Federal Reserve Bank of St. Louis. Potential output is constructed by the Congressional Budget Office. Periods of recession are shaded, using the dates assigned by the National Bureau of Economic Research.
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Short Run

- Typically, during a recession, output is below potential for approximately 2-4 years
 - ▶ About 6 percent of GDP is foregone
 - ▶ Loss of about \$3,000 per person and between 1.5 million and 3 million jobs are lost
- In addition, the decline in income is typically not spread evenly across the population
 - ▶ To the extent that it's concentrated in particular regions, industries, and families, the costs to those affected can be even larger
 - ▶ Unemployment rate rises by one or two percentage points
- For a complete analysis of a cycle, we may also incorporate the benefits of a boom

Changes in U.S. Employment

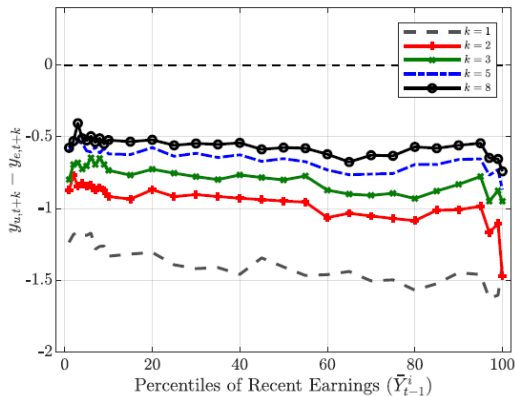
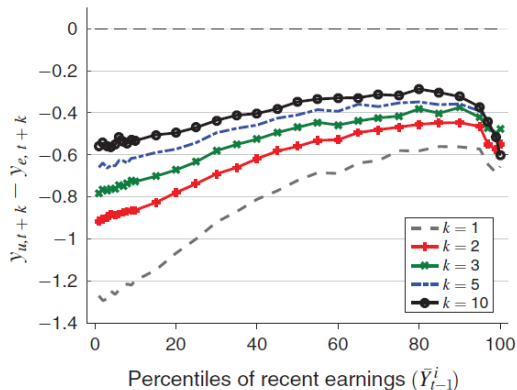
- Notice how severe the recent recession appears
- Total employment fell by more than 8.2 million jobs between Dec 2007 and Dec 2009: this is the largest decrease in employment in several decades



Source: World Bank, Inflation, consumer prices for the United States [FPCPITOTLZGUSA], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/FPCPITOTLZGUSA>, September 23, 2019.
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Scarring Effect

- Below shows earning differences for workers who were jobless for over a year compared to those with similar income levels in the past five years by percentiles
- Left (right) panel shows the difference in average income after k years for the US (Canada)



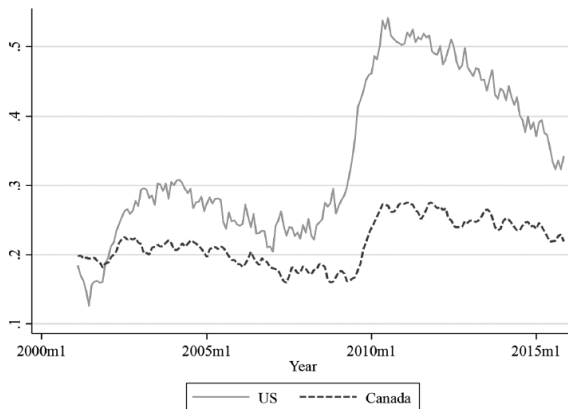
⁰Source: Guvenen et.al (2017) and Kurnaz et.al (2024)

Scarring Effect

- Canada experienced significantly higher income losses over the five years following post-unemployment
 - ▶ Losses in the US range between 40-60 log points, whereas Canadian losses range between 60-80 log points
- Losses in Canada demonstrate a weakly increasing trend over the income distribution, in contrast to the inverted U-shaped pattern observed in the US
- The challenges of finding employment during a recession may lead to long-term negative effects for the unemployed beyond the initial 1-2 years of hardship

Long-term Unemployment in the US and Canada

- Relative to the United States, Canada experiences a more modest increase in long-term unemployment.
- In both countries, long-term unemployment persists for a period exceeding 5 years
 - ▶ Long-term unemployment is defined as being unemployed for roughly half a years



⁰Source: Kroft et.al (2019)

Measuring Potential Output

- There is no directly observable measure of potential output in an economy
- Ways to measure potential output:
 - ▶ Fitting perfectly smooth trend passes through quarterly movements of real GDP
 - ★ e.g., Hodrick-Prescott filter, Baxter-King filter and more recently, Hamilton Filter
 - ▶ Take averages of the surrounding actual GDP numbers
 - ★ e.g., Moving average
- Annualized rate: rate of change that would apply if the growth rate persisted for an entire year
 - ▶ If GDP increases by 1 percent over a single quarter, it will have grown at an annualized rate of 4 percent

Measuring Potential Output

- Suppose we observe that GDP grew over the last quarter at an annualized rate of 5 percent
 - ▶ How much of this increase in GDP reflects an increase in potential output?
 - ▶ How much reflects a short-run fluctuation?
 - ▶ Is the economy now above potential, or has potential output grown rapidly?

Measuring Potential Output

- Economists consult a variety of indicators to answer this question. They examine
 - ▶ data on how quickly the unemployed are finding jobs
 - ▶ surveys revealing economic activity in particular industries
 - ▶ demand by firms for new factories and equipment, and so on
- If policymakers miscalculate whether the economy is in a recession or expansion, this can lead to unintended consequences, and sometimes exacerbate the effects of the business cycle

The Short-Run Model

The short-run model is based on three premises:

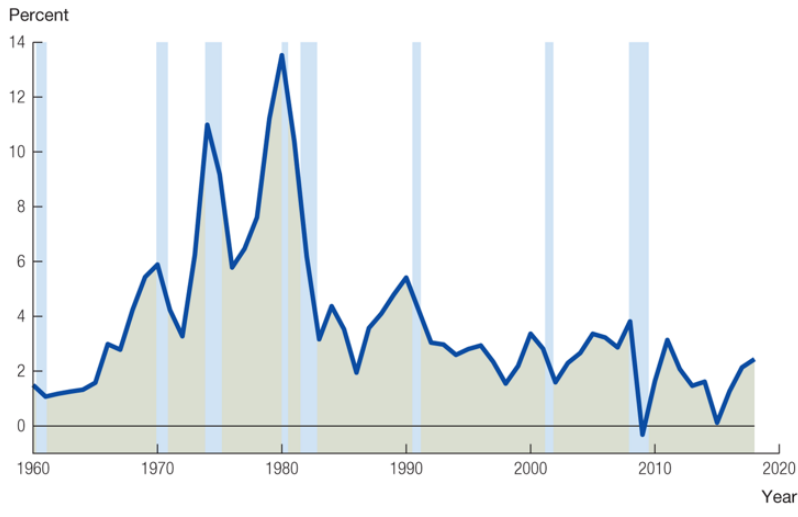
- The economy is constantly being hit by shocks
 - ▶ Shocks: factors that cause fluctuations or even the potential output
 - ★ e.g., changes in oil prices, disruptions in financial markets, development of new technologies, natural disasters, etc
- Monetary and fiscal policies affect output
 - ▶ Policymakers can neutralize shocks to the economy
 - ▶ e.g., BoC lowered the interest rate in the beginning of the pandemic
- There is a dynamic trade-off between output and inflation shown by the Phillips curve

The Phillips Curve (PC)

- The Phillips curve (PC) shows
 - ▶ a positive relationship between the change in inflation and short-run output
 - ▶ a boom increases inflation
 - ▶ a recession decreases inflation
- Note we are interested in the change in inflation $\Delta\pi$ not the inflation rate itself π
 - ▶ Traditionally, the PC focus on the inflation rate
 - ▶ In 60s-70s, Milton Friedman (1968) and Edmund Phelps enhance the PC by introducing expectation to PC (more on that later)

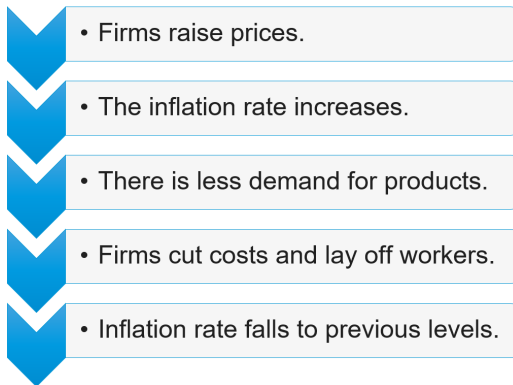
The Inflation Rate

- The rate of inflation typically
 - ▶ peaks at the start of a recession
 - ▶ falls during the recession



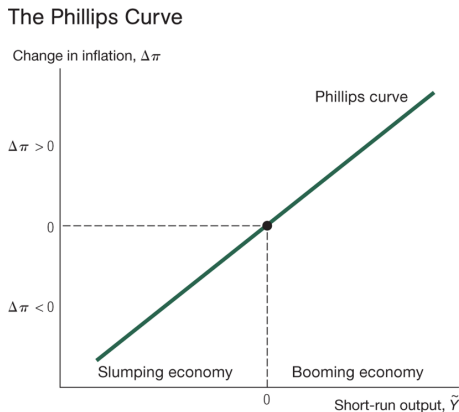
Works in a "Cycle"

- Prices rise faster than the inflation rate for two reasons:
 - ▶ In the economic boom, demand is higher, all firms are producing above potential, so higher prices follow
 - ★ the firms must raise wages, and perhaps delay maintenance on machines
 - ★ Firms must raise prices above the prevailing inflation rate to cover these costs



The Phillips Curve

- The Phillips curve relates the change in the inflation rate to the amount of economic activity
 - ▶ In a booming economy ($\tilde{Y} > 0$), inflation rises ($\Delta\pi > 0$)
 - ▶ In a slumping economy ($\tilde{Y} < 0$), inflation falls ($\Delta\pi < 0$)



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Measuring the Phillips Curve

- The positive sloped PC fit the data well for long time horizon



Source: FRED. U.S. Department of Commerce.
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Measuring the Phillips Curve

- Empirically, the slope is about $1/3$
 - ▶ If output exceeds potential by 3 percent, the inflation rate increases one percentage point
- In year 1979, inflation was high due to oil prices shock. The central bank decided to rise the interest rate to lower the inflation
 - ▶ As learn from the consumption/investment model, as interest increase, investment and consumption would fall. Thus, output will fall
- We can see that the points on the graph from 79-82, it is moving down along the PC

Okun's Law: Output and Unemployment

- Natural rate of unemployment: the rate of unemployment that exists in the long run, defined by $\bar{u}$
- Cyclical unemployment is the current unemployment minus the natural rate of unemployment

$$u - \bar{u}$$

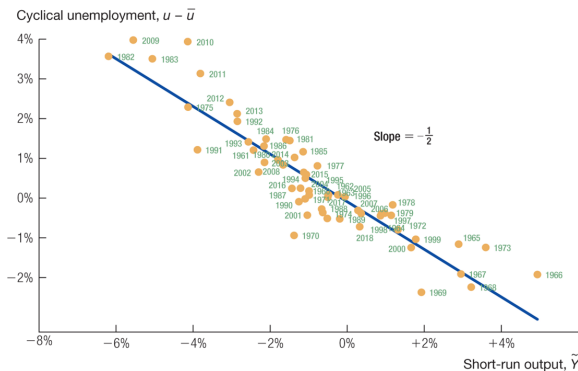
- In a recession, output is low and thus unemployment is high
- The relationship between these two variables is known as Okun's law

$$u - \bar{u} = -\frac{1}{2} \times \tilde{Y}$$

- Okun's law says that for each percentage point that output is below potential, the unemployment rate exceeds its long-run level by half a percentage point

Okun's Law for the U.S. Economy

- Variable $\bar{u}$ is measured as the average rate in the surrounding period
- There is a tight, negative relationship between output and unemployment



Source: FRED. U.S. Department of Commerce.
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Next Lecture

- Next lecture will start Ch.11